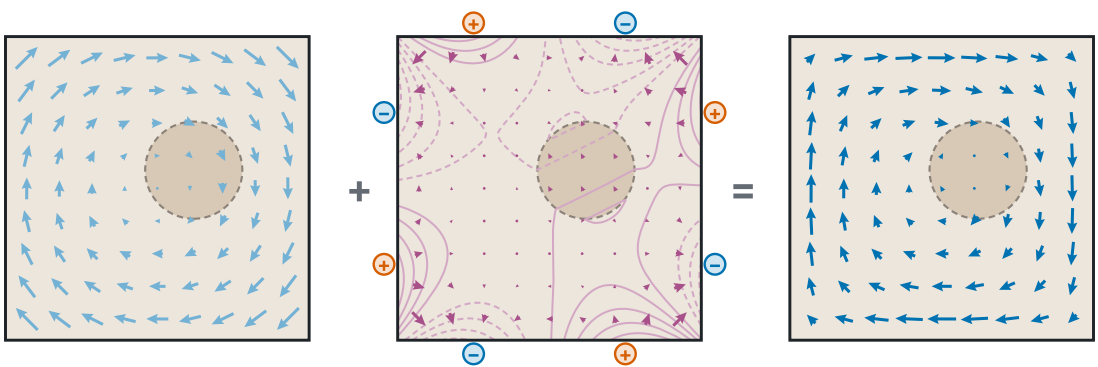

primary $\mathbf{E}_p = -\partial_t \mathbf{A}$
correction $-\nabla\phi$
total $\mathbf{E} = \mathbf{E}_p - \nabla\phi$
+
=
pushes current out
from charges; curl-free
runs along the boundary